

**Department of Computer Science & Engineering (IoT)****Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.***Session 2026-2027****Vision:** Dream of where you want.**Mission:** Means to achieve Vision**Program Educational Objectives of the program (PEO):** (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)**Keywords of POs:**

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name of Student

**Department of Computer Science & Engineering (IoT)****Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.*

Session	2026-27 (ODD)	Course Name	Lab : Embeded Systems
Semester	7	Course Code	23IOT1706
Roll No	61	Name of Student	Sheikh Wasimuddin

Practical Number	05
Course Outcome	CO1: Explain the fundamentals of embedded microcontrollers, development environments, and simulation tools. CO2: Develop embedded applications using Arduino/ESP32 and interface sensors, actuators, and peripherals. CO3: Implement event-driven and communication-based embedded systems using interrupts, UART, and I2C protocols. CO4: Design, simulate, and test embedded system applications for real-world applications.
Aim	Understand master-slave communication between two Arduino boards using I2C (Inter-Integrated Circuit) protocol.
Theory (100 words)	I2C (Inter-Integrated Circuit) is a synchronous serial communication protocol commonly used in embedded systems for communication between a microcontroller and multiple peripheral devices. It uses only two communication lines: SDA (Serial Data Line) for transferring data and SCL (Serial Clock Line) for synchronizing the communication. I2C follows a Master-Slave communication architecture. The Master device controls the communication and generates the clock signal, while the Slave device responds to commands from the Master. Each Slave is identified using a unique I2C address, which allows multiple devices to operate on the same bus. In this experiment, two Arduino Uno boards are used to demonstrate I2C communication. One Arduino is configured as the Master and the other as the Slave. A push button is connected to the Master Arduino to provide an input. The Master reads the button state and sends the corresponding data to the Slave through the I2C bus. The Slave Arduino receives the data using its assigned I2C address. According to the received value, it controls an LED. When the button on the Master is pressed, the Master sends a signal to the Slave, causing the LED to turn ON. When the button is released, the Slave receives the corresponding

**Department of Computer Science & Engineering (IoT)****Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.*

	signal and turns the LED OFF. On the Arduino Uno, the I2C communication lines are A4 (SDA) and A5 (SCL). The Wire library is used to establish I2C communication and provides functions for starting the bus, transmitting data, receiving data, and handling communication events. The major advantage of I2C is that multiple devices can share the same two communication lines, reducing the number of wires required. It is widely used for interfacing sensors, LCD displays, EEPROMs, RTC modules, and other peripheral devices with microcontrollers.
Procedure and Execution (100 Words)	1. Open Wokwi/Tinkercad Circuits and create a new circuit. 2. Add two Arduino Uno boards to the circuit. 3. Configure the first Arduino as the I2C Master and the second Arduino as the I2C Slave. 4. Connect a push button between D2 and GND of the Master Arduino. 5. Connect an LED to D13 of the Slave Arduino. The built-in LED on pin D13 can also be used. 6. Connect A4 (SDA) of the Master Arduino to A4 (SDA) of the Slave Arduino. 7. Connect A5 (SCL) of the Master Arduino to A5 (SCL) of the Slave Arduino. 8. Connect the GND of both Arduino boards together. 9. Open the Arduino IDE/editor and write the Master program using the Wire library. 10. Set the push-button pin as an input using the internal pull-up resistor. 11. Configure the Master to send the button status to the Slave using the Slave's I2C address. 12. Write the Slave program and assign it a unique I2C address, such as 8. 13. Configure the Slave to receive data using the Wire.onReceive() function. 14. Upload/run the Master and Slave programs on their respective Arduino boards. 15. Start the simulation and press the push button connected to

Department of Computer Science & Engineering (IoT)**Vision of the Department***To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.***Mission of the Department***To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.*

the Master.

16.Observe that the Master sends the button status to the Slave through SDA and SCL.

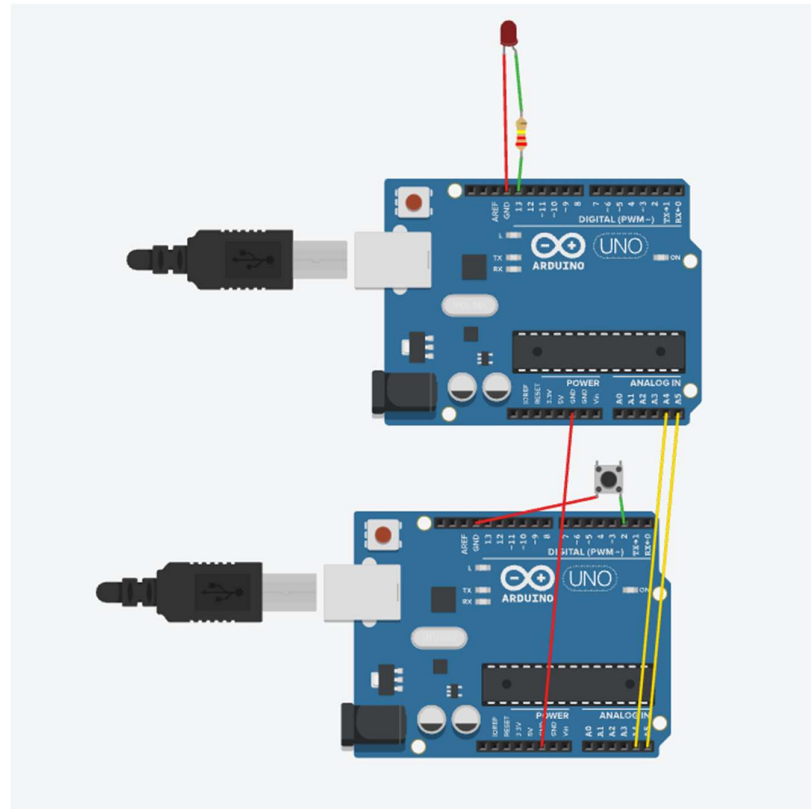
17.Verify that the Slave turns the LED ON when the button is pressed.

18.Release the push button and verify that the LED turns OFF.

19.Repeat the experiment several times to confirm reliable Master-Slave communication.

20.Record the observations and verify the successful implementation of I2C communication between two Arduino Uno boards.

Stepwise Screenshots with steps:



Department of Computer Science & Engineering (IoT)

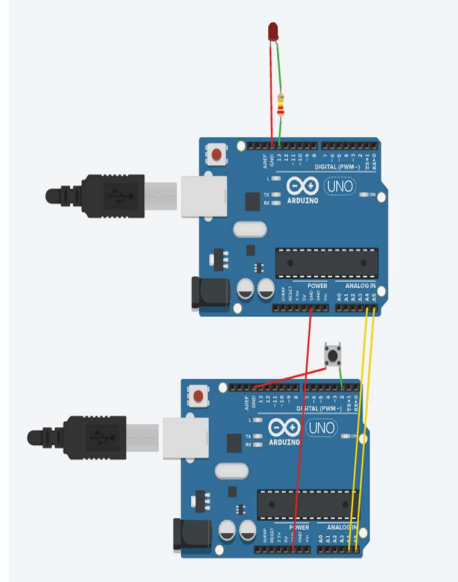
Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

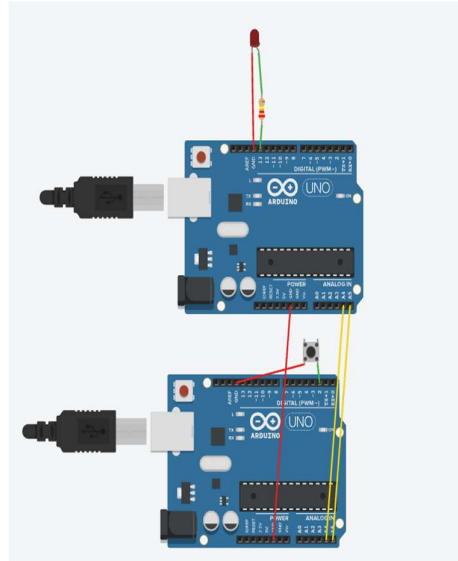
To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.

Code of 1 Arduino



```
1 //sheikhwasimuddin 61 ES LAB5
2 #include <Wire.h>
3
4 #define SLAVE_ADDRESS 8
5 #define BUTTON_PIN 2
6
7 void setup() {
8   Wire.begin();           // Start I2C as Master
9   pinMode(BUTTON_PIN, INPUT_PULLUP);
10 }
11
12 void loop() {
13   int buttonState = digitalRead(BUTTON_PIN);
14
15   Wire.beginTransmission(SLAVE_ADDRESS);
16
17   if (buttonState == LOW) {
18     Wire.write(1);         // Button pressed
19   } else {
20     Wire.write(0);         // Button released
21   }
22
23   Wire.endTransmission();
24
25   delay(100);
26 }
```

Code of 2 Arduino



```
1 //sheikhwasimuddin 61 ES LAB5
2 #include <Wire.h>
3
4 #define SLAVE_ADDRESS 8
5 #define LED_PIN 13
6
7 void setup() {
8   pinMode(LED_PIN, OUTPUT);
9
10   Wire.begin(SLAVE_ADDRESS); // Start I2C as Slave
11   Wire.onReceive(receiveData); // Register receive function
12 }
13
14 void loop() {
15   // Main loop remains free
16   delay(10);
17 }
18
19 void receiveData(int bytes) {
20   if (Wire.available()) {
21     int data = Wire.read();
22
23     if (data == 1) {
24       digitalWrite(LED_PIN, HIGH);
25     }
26     else {
27       digitalWrite(LED_PIN, LOW);
28     }
29   }
30 }
```

Department of Computer Science & Engineering (IoT)

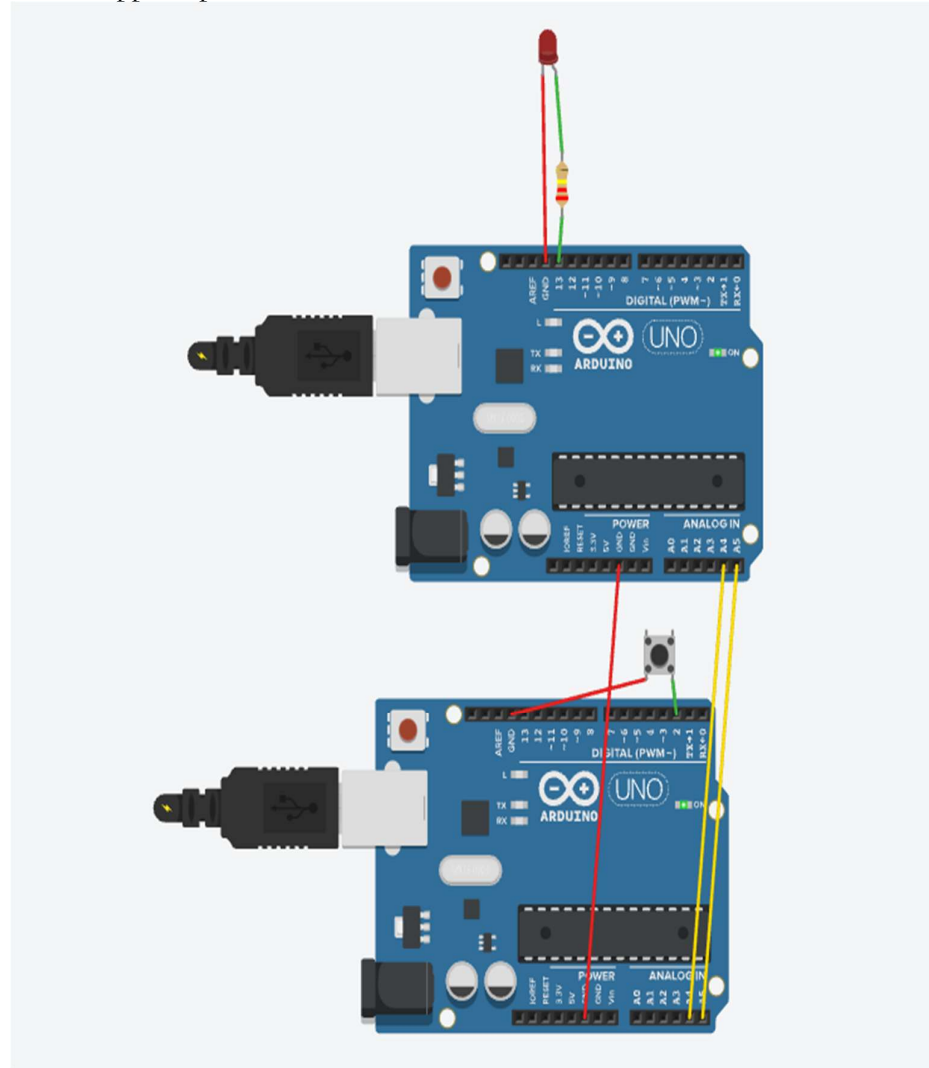
Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.

When supplied power



Conclusion

The I2C communication between two Arduino Uno boards was successfully implemented. One Arduino was configured as the **Master** to read the push-button status and transmit the data through the **SDA and SCL** lines. The second Arduino was configured as the **Slave** to receive the data and control the LED accordingly. The LED turned **ON** when the button was pressed and **OFF** when the button was released. Thus, the experiment successfully demonstrated **Master-Slave communication using the I2C protocol with only two communication lines**.

Date

20-08-2026